

MSc

Course title: Bayes Methods

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1. Let $A = [0, 1]^2$ be the unit square region of $\mathbb{R}^2$. For $i = 1, \dots, m$, let $y_i = (y_{i,1}, y_{i,2})$ be points in A and let $y = \{y_1, \dots, y_m\}$ be a set of m points. The number and locations of the points are distributed according to a Strauss point process with parameters $\lambda > 0, 0 \leq \gamma \leq 1$ and $R \geq 0$. In particular, let $c(y; R)$ be the function

$$c(y; R) = \sum_{i=1}^{m-1} \sum_{j=i+1}^m \mathbb{I}_{|y_i - y_j| < R}$$

counting the number of R -close pairs of points in y . Let Ω_m be the set of all distinguishable sets of m points $y = \{y_1, \dots, y_m\}$ in $y \subset A$ and let $\Omega = \cup_{m=0}^{\infty} \Omega_m$. The density of $y|\lambda, \gamma, R$ is

$$p(y|\lambda, \gamma, R) = \xi(\lambda, \gamma, R) \gamma^{c(y, R)} \lambda^m,$$

where $\xi(\lambda, \gamma, R)$ is a normalising constant ensuring the density is normalised to one over Ω .

- (a) [8 marks] Suppose y is observed and we wish to estimate λ, γ, R .
- (i) Write down the posterior distribution $\pi(\lambda, \gamma, R|y)$ in as much detail as you can, but without specifying priors for λ, γ, R .
 - (ii) Explain why $\pi(\lambda, \gamma, R|y)$ is doubly intractable.
 - (iii) Suppose an algorithm simulating $x \sim p(x|\lambda, \gamma, R)$ is available. Give an ABC algorithm approximately targeting $\pi(\lambda, \gamma, R|y)$ (state what summary statistics and distance measure you are using).
- (b) [12 marks] Suppose now λ, γ and R are known, but the point pattern y is in fact a thinned realisation of a Strauss process. The true underlying set of points $z = \{z_1, \dots, z_n\}, z \in \Omega, n \geq m$ has a Strauss distribution in $A = [0, 1]^2$,

$$p(z|\lambda, \gamma, R) \propto \gamma^{c(z, R)} \lambda^n.$$

For $i = 1, \dots, n$, point z_i is observed independently, so $z_i \in y$, with probability q , and otherwise z_i is missed. In that case $z_i \notin y$. Here q , the probability a point gets observed, is unknown with prior $\pi(q)$ independent of z . We wish to estimate a credible interval for q .

- (i) Let z be a point pattern of n points satisfying $y \subseteq z$. Calculate $\pi(z, q|y)$
- (ii) Give a reversible jump MCMC algorithm targeting $\pi(z, q|y)$. Your algorithm should contain updates for adding and deleting points $(x_1, x_2) \in z \setminus y$, and an update for q . Give the acceptance probabilities as explicitly as you can.
- (iii) Explain how to compute a $100(1 - \alpha)\%$ equal tailed credible interval for q from MCMC output $q^{(t)}, t = 1, \dots, T$. What is the relationship to model averaging here?

2. (a) [5 marks] A random distribution $G \sim DP(\alpha, H)$ on Ω has a Dirichlet Process distribution with parameter α and base distribution H on Ω .
- Give a definition of a Dirichlet Process.
 - Calculate the expectation and variance of $G(A)$, $A \subseteq \Omega$ in terms of α and $H(A)$, and briefly interpret the role α and H play in determining the distribution of G .

$$\text{Hint: If } X \sim \text{Beta}(a, b), \text{ then } E(X) = \frac{a}{a+b}, \text{ and } \text{var}(X) = \frac{ab}{(a+b)^2(a+b+1)}.$$

- (b) [9 marks] Consider regression with a Dirichlet-Normal mixture for the error distribution. Let $G \sim DP(\alpha, H)$ with $H(d\theta_i) = N(d\theta_i; 0, \sigma_\theta^2)$. Here $\theta_i \in \mathfrak{R}$ so $\Omega = \mathfrak{R}$. For $i = 1, \dots, n$ let x_i, y_i be a scalar covariate and response and β a scalar effect. The regression is

$$y_i = \theta_i + x_i\beta + \epsilon_i, \quad \text{with } \epsilon_i \sim N(0, \sigma_\epsilon^2) \text{ and } \theta_i \sim G$$

independently for each $i = 1, \dots, n$. Let $\theta = (\theta_1, \dots, \theta_n)$ and $\sigma = (\sigma_\epsilon, \sigma_\theta)$.

- Let $\theta^* = (\theta_1^*, \dots, \theta_K^*)$ list the distinct values in θ . Let $S = (S_1, \dots, S_K)$ be the partition of $\{1, 2, \dots, n\}$ satisfying $i \in S_k$ if $\theta_i = \theta_k^*$ and let $n_k = \text{card}(S_k)$. Let $P(S)$ be the probability θ has partition S .

Assume a prior $\pi(\beta, \sigma)$ is available. Write down the posterior for β, σ, θ^* and S in as much detail as you can (α is known and fixed).

$$\text{Hint: You may use without proof that } P(S) = \frac{\alpha^K \prod_{k=1}^K (n_k - 1)!}{\prod_{i=1}^n (\alpha + i - 1)}.$$

- Let $K^{-i}, S^{-i} = (S_1^{-i}, \dots, S_{K^{-i}}^{-i})$ and θ^{*-i} correspond to $K, S = (S_1, \dots, S_K)$ and θ^* after i is removed from its partition set (but note that y_i is retained). Calculate $\Pr(i \in S_k | S^{-i}, \beta, \sigma, \theta^{*-i}, y)$, the conditional distribution for the partition for the i 'th response, given the other parameters and the partition for the rest of the data and briefly indicate why this distribution is useful as part of an MCMC algorithm targeting $\pi(\beta, \sigma, \theta^*, S | y)$.

- (c) [6 marks]

- Calculate a Laplace approximation to $\log(p(y|S))$, where $p(y|S)$ is the marginal likelihood for the data y given a partition S .

Hint: consider $z = (\beta, \sigma, \theta^)$ in the Laplace approximation,*

$$\int e^{-nh(z)} dz = e^{nh(\hat{z})} (2\pi/n)^{p/2} |\Sigma|^{1/2} + O(n^{-3/2})$$

where $h(z)$ is a given smooth function of $z = (z_1, \dots, z_p)$, $\Sigma^{-1} = \frac{\partial^2 h}{\partial z \partial z^T} \Big|_{z=\hat{z}}$ is the Hessian of $h(z)$ at $\hat{z}$ with determinant $|\Sigma|^{-1}$, and $\hat{z}$ maximises $h(z)$.

- In a problem with n small it is feasible to estimate $p(y|S)$ for every possible partition S using a Laplace approximation. A researcher interested in estimating the partition does this and reports the partition with the largest estimated value of $p(y|S)$ as a best estimate for the unknown true partition. What criticisms do you have of this procedure?